3 **(a)** Explain what is meant by simple harmonic motion.

.....

.....

.....

..... [2]

(b) Fig. 3.1 shows the variation with time t of the height h above the ground of an object of mass 37 kg that is undergoing vertical simple harmonic motion.

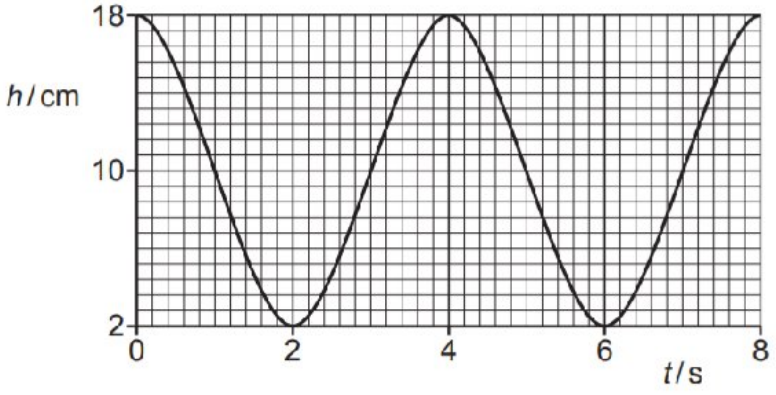


Fig. 3.1

For the oscillations of the object,

(i) determine the amplitude x_0,

x_0 = ... cm [1]

(ii) show that the angular frequency is 1.6 rad s^{-1},

[1]

(iii) determine the total energy E.

E = ... J [2]

(c) On Fig. 3.2, sketch the variation with time t of the kinetic energy E_K of the object.

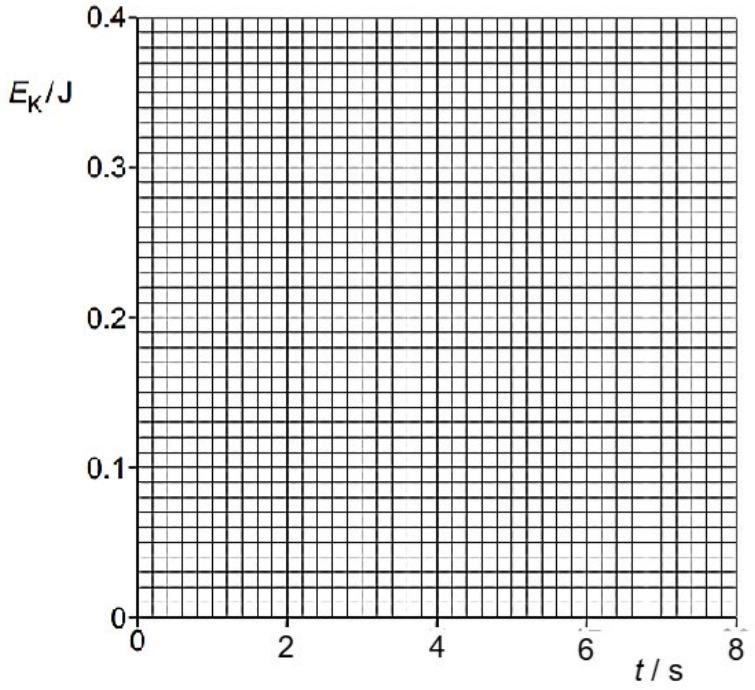


Fig. 3.2

[2]

(d) On Fig. 3.3, sketch the variation with the height h above the ground of the kinetic energy E_K of the object.

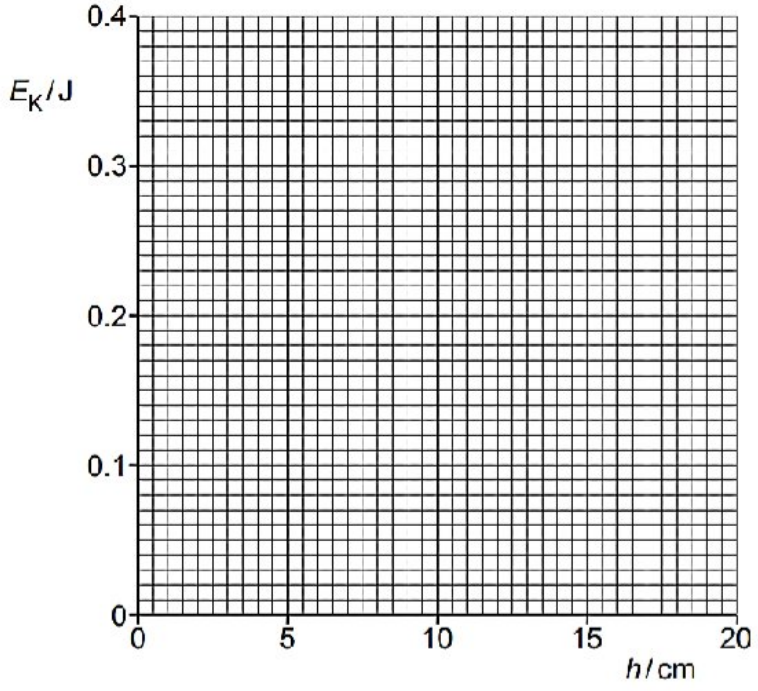


Fig. 3.3

[2]

[Total: 10]